Orthogonality

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Orthogonality decomposes vectors and subspaces. Projections lead to orthogonal bases and least squares; the four fundamental subspaces describe the complete geometry of $Ax = b$.

Orthogonal Complements

$$V \perp W \iff v^T w = 0 \,\forall v \in V, w \in W, \qquad S^{\perp} = \{x : x^T s = 0 \,\forall s \in S\}.$$

$$\mathcal{R}(A) = \mathcal{N}(A)^{\perp}, \quad \mathcal{N}(A) = \mathcal{R}(A)^{\perp}, \quad \mathcal{C}(A) = \mathcal{N}(A^T)^{\perp}.$$

$$\mathbb{R}^n = \mathcal{R}(A) \oplus \mathcal{N}(A), \qquad x = x_r + x_n.$$

Projections

Projection onto a line:
$\operatorname{proj}_a b = \dfrac{a^T b}{a^T a} a.$

$$b = p + e, \quad p = \operatorname{proj}_a b, \quad e \perp a, \quad \|b\|^2 = \|p\|^2 + \|e\|^2.$$

Projection matrix:
$P = \frac{aa^T}{\|a\|^2}$, $P^2 = P$, $P^T = P$.

Orthonormal columns:
$Q^T Q = I$, and projection onto $\mathcal{C}(Q)$ is $P = QQ^T$.

Least Squares

Normal equations:
$A^T A\hat{x} = A^T b$; residual $e = b - A\hat{x}$ satisfies $A^T e = 0$.

$$\hat{x} = (A^T A)^{-1} A^T b, \qquad p = A\hat{x} = Pb.$$

Gram–Schmidt:
$q_1 = a_1/\|a_1\|$; $v_k = a_k - \displaystyle\sum_{j<k} (q_j^T a_k) q_j$, $q_k = v_k/\|v_k\|$.

QR factorization:
$A = QR$, where $Q^T Q = I$ and R is upper triangular.